INCH-POUND

MIL-M-28787/508A
AMENDMENT 1
3 August 1994

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

CENTRAL PROCESSOR SEQUENCER MODULE, KEY CODE BWAC

This amendment forms a part of Military Specification MIL-M-28787/508A, dated 23 March 1994, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGES 4 THROUGH 8

TABLE I: Delete and substitute new table I as printed on pages 2 through 5 of this amendment.

PAGES 21 THROUGH 26

6.3.2, Output parameters, Ios max (mA) column: Delete Ios max (mA) column.

PAGES 53 THROUGH 58

TABLE VI: Delete and substitute new table VI as printed on pages 6 through 10 of this amendment.

Custodians:
Army - ER
Navy - NW
Air Force - 85

Review activities:
Army - AT
Navy - MC
Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - NW

(Project 5963-0863)

AMSC N/A
FSC 5963
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:24Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements.

Test cond.	Parameter 5/	Ref test table	Test type 1/ - 3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 80 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
A	Isolation	II	S	10	INF	10	INF	10	INF	MOhm
	End-of-life	II	S	1	INF	1	INF	1	INF	MOhm
B1-B6	Continuity	III	X	0	1	0	1	0	1	Ohm
C	Vcc stress	IV	S	-------Module shall pass all remaining tests------						
D1,D2	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D3-D5	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D6	Ibd	V	S	0.0	2.3	0.0	2.3	0.0	2.3	mA
D7	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D8	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D9	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D10,D11	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D12	Ibd	V	S	0.0	1.1	0.0	1.1	0.0	1.1	mA
D13	Ibd	V	S	-0.02	2.12	-0.02	2.12	-0.02	2.12	mA
D14	Ibd	V	S	0.0	1.1	0.0	1.1	0.0	1.1	mA
D15	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D16	Ibd	V	S	-0.02	2.12	-0.02	2.12	-0.02	2.12	mA
D17	Ibd	V	S	0.0	2.0	0.0	2.0	0.0	2.0	mA
D18	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D19	Ibd	V	S	0.0	3.0	0.0	3.0	0.0	3.0	mA
E1-E73	Ios	VI	S	-------Module shall pass all remaining tests------						
F1,F2	Iih	VII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
F3-F5	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F6	Iih	VII	X	-5.8	0.16	-5.8	0.16	-5.8	0.16	mA
F7	Iih	VII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
F8,F9	Iih	VII	X	-0.8	0.24	-0.8	0.24	-0.8	0.24	mA
F10	Iih	VII	X	-3.65	0.21	-3.65	0.21	-3.65	0.21	mA

- See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:24Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements - Continued.

Test cond.	Parameter 5/	Ref test table	Test type 1/ - 3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
F11	Iih	VII	X	-3.67	0.23	-3.67	0.23	-3.67	0.23	mA
F12	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F13	Iih	VII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
F14	Iih	VII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
F15	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F16-F25	Iih	VII	X	-3.65	0.21	-3.65	0.21	-3.65	0.21	mA
F26	Iih	VII	X	-2.4	0.07	-2.4	0.07	-2.4	0.07	mA
F27	Iih	VII	X	-3.11	-0.52	-3.11	-0.52	-3.11	-0.52	mA
F28	Iih	VII	X	-2.6	0.07	-2.6	0.07	-2.6	0.07	mA
F29	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F30	Iih	VII	X	-5.75	-0.49	-5.75	-0.49	-5.75	-0.49	mA
F31	Iih	VII	X	-4.0	0.1	-4.0	0.1	-4.0	0.1	mA
F32	Iih	VII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
F33	Iih	VII	X	-6.0	0.15	-6.0	0.15	-6.0	0.15	mA
G1,G2	Iil	VIII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
G3-G5	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G6	Iil	VIII	X	-5.8	0.16	-5.8	0.16	-5.8	0.16	mA
G7	Iil	VIII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
G8,G9	Iil	VIII	X	-0.8	0.24	-0.8	0.24	-0.8	0.24	mA
G10	Iil	VIII	X	-3.65	0.21	-3.65	0.21	-3.65	0.21	mA
G11	Iil	VIII	X	-3.67	0.23	-3.67	0.23	-3.67	0.23	mA
G12	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G13	Iil	VIII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
G14	Iil	VIII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
G15	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G16-G25	Iil	VIII	X	-3.65	0.21	-3.65	0.21	-3.65	0.21	mA
G26	Iil	VIII	X	-2.4	0.07	-2.4	0.07	-2.4	0.07	mA
G27	Iil	VIII	X	-3.11	-0.52	-3.11	-0.52	-3.11	-0.52	mA
G28	Iil	VIII	X	-2.6	0.07	-2.6	0.07	-2.6	0.07	mA

- See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:24Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements - Continued.

Test cond.	Parameter 5/	Ref test table	Test type 1/ - 3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
G29	III	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G30	III	VIII	X	-5.75	-0.49	-5.75	-0.49	-5.75	-0.49	mA
G31	III	VIII	X	-4.0	0.1	-4.0	0.1	-4.0	0.1	mA
G32	III	VIII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
G33	III	VIII	X	-6.0	0.15	-6.0	0.15	-6.0	0.15	mA
H1-H73	Voh	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J74	Vol	X	X	0.0	0.5	0.0	0.5	0.0	0.5	V
L1-L9	Vic	XI	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L10	Vic	XI	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L11-L17	Vic	XI	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
M1	PSFC	XII	S	4.9	9.1	4.9	9.1	4.9	9.1	µF
M2	PSFC	XII	S	5.6	10.4	5.6	10.4	5.6	10.4	µF
N	Icc	XIII	X	630	2544	630	2544	630	2544	mA
P1,P2	Functional	XIV	X	See table XIV						
Q1,Q2	Dynamic	XV	S	See table XV						
Q3,Q4	Dynamic	XV	Q	See table XV						
R1-R5	CI/o	XVII	Q	2	50	2	50	2	50	pF
R6	CI/o	XVII	Q	10	100	10	100	10	100	pF
R7-R11	CI/o	XVII	Q	2	50	2	50	2	50	pF
R12	CI/o	XVII	Q	4	75	4	75	4	75	pF
R13	CI/o	XVII	Q	8	175	8	175	8	175	pF
R14	CI/o	XVII	Q	4	100	4	100	4	100	pF
R15	CI/o	XVII	Q	2	50	2	50	2	50	pF
R16	CI/o	XVII	Q	8	125	8	125	8	125	pF
R17	CI/o	XVII	Q	4	75	4	75	4	75	pF
R18	CI/o	XVII	Q	2	75	2	75	2	75	pF
R19	CI/o	XVII	Q	6	100	6	100	6	100	pF

- See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:24Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements - Continued.

Test cond.	Parameter 5/	Ref test table	Test type 1/ - 3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
R20	Ci/o	XVII	Q	16	321	16	321	16	321	pF
R21	Ci/o	XVII	Q	16	312	16	312	16	312	pF
R22	Ci/o	XVII	Q	14	364	14	364	14	364	pF
R23	Ci/o	XVII	Q	14	200	14	200	14	200	pF
R24	Ci/o	XVII	Q	14	273	14	273	14	273	pF
R25	Ci/o	XVII	Q	14	263	14	263	14	263	pF
R26	Ci/o	XVII	Q	14	262	14	262	14	262	pF
R27	Ci/o	XVII	Q	14	254	14	254	14	254	pF
R28	Ci/o	XVII	Q	14	261	14	261	14	261	pF
R29	Ci/o	XVII	Q	14	283	14	283	14	283	pF
R30	Ci/o	XVII	Q	14	279	14	279	14	279	pF
R31	Ci/o	XVII	Q	14	328	14	328	14	328	pF
R32,R33	Ci/o	XVII	Q	2	50	2	50	2	50	pF
R34,R35	Ci/o	XVII	Q	4	75	4	75	4	75	pF
R36	Ci/o	XVII	Q	4	100	4	100	4	100	pF
R37	Ci/o	XVII	Q	12	204	12	204	12	204	pF
R38	Ci/o	XVII	Q	10	150	10	150	10	150	pF
R39	Ci/o	XVII	Q	8	175	8	175	8	175	pF
R40	Ci/o	XVII	Q	8	200	8	200	8	200	pF
R41-R47	Ci/o	XVII	Q	2	50	2	50	2	50	pF
R48-R64	Ci/o	XVII	Q	4	75	4	75	4	75	pF
R65	Ci/o	XVII	Q	2	75	2	75	2	75	pF
R66	Ci/o	XVII	Q	2	50	2	50	2	50	pF
R67	Ci/o	XVII	Q	2	75	2	75	2	75	pF
R68	Ci/o	XVII	Q	4	150	4	150	4	150	pF
R69	Ci/o	XVII	Q	6	200	6	200	6	200	pF
R70	Ci/o	XVII	Q	4	75	4	75	4	75	pF
R71	Ci/o	XVII	Q	2	75	2	75	2	75	pF
R72-R75	Ci/o	XVII	Q	4	75	4	75	4	75	pF
R76	Ci/o	XVII	Q	2	75	2	75	2	75	pF
T1,T2	PSTA	XVIII	Q	0	400	0	400	0	400	mV

- See footnotes on next page.

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:24Z
Check the source to verify that this is the current version before use.

TABLE VI. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits	Notes
E1	EM-&-INT-PND	2	0.0	Module shall pass all remaining tests.	1-4
E2	RESET-Y-FULL	4			
E3	RST-MDRI-FUL	6			
E4	SMI-D-IN-0	11			
E5	WAIT-REQ	13			
E6	READY	14			
E7	PROM-47	16			
E8	PROM-47	16			
E9	PROM-46	17			
E10	PROM-46	17			
E11	MCRO-P-10	18			
E12	MCRO-P-11	19			
E13	WAIT-ENABLE	22			
E14	SBEQADDGENSC	23			
E15	SB-EQ-ECI-SC	24			
E16	MCROIR-45	27			
E17	MCROIR-44	28			
E18	MCROIR-43	29			
E19	MCROIR-41	32			
E20	MCROIR-40	33	0.0	Module shall pass all remaining tests.	1-4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:24Z
Check the source to verify that this is the current version before use.

TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits	Notes
E21	XPANDED-ADRS	34	0.0	Module shall pass all remaining tests.	1-4
E22	MCROIR-38	35			
E23	MCROIR-37	36			
E24	MCROIR-36	37			
E25	MCROIR-35	38			
E26	MCROIR-34	39			
E27	MCROIR-33	41			
E28	MCROIR-32	42			
E29	MCROIR-31	43			
E30	MCROIR-30	44			
E31	MCROIR-29	45			
E32	MCROIR-28	46			
E33	MCROIR-27	47			
E34	MCROIR-26	48			
E35	MCROIR-25	49			
E36	MCROIR-24	50			
E37	MICRO-P00	52			
E38	MICRO-P01	53			
E39	MICRO-P02	54			
E40	MICRO-P03	55	0.0	Module shall pass all remaining tests.	1-4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:24Z
Check the source to verify that this is the current version before use.

TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits	Notes
E41	MICRO-P04	56	0.0	Module shall pass all remaining tests.	1-4
E42	MICRO-P05	57			
E43	MICRO-P06	58			
E44	MICRO-P07	59			
E45	MICRO-P08	61			
E46	MICRO-P09	62			
E47	MCROIR-00	65			
E48	MCROIR-01	66			
E49	MCROIR-02	67			
E50	MCROIR-03	68			
E51	MCROIR-04	69			
E52	MCROIR-05	70			
E53	MCROIR-06	71			
E54	MCROIR-07	72			
E55	MCROIR-08	73			
E56	MCROIR-09	74			
E57	MCROIR-10	77			
E58	MCROIR-11	78			
E59	MCRO-CNT-FMT	79			
E60	ECI-DESTNS	82	0.0	Module shall pass all remaining tests.	1-4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:24Z
Check the source to verify that this is the current version before use.

TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits	Notes
E61	EMULATE-OE	86	0.0	Module shall pass all remaining tests.	1-4
E62	MCROIR-12	87			
E63	MCROIR-13	88			
E64	MCROIR-14	91			
E65	MCROIR-15	92			
E66	MCROIR-16	93			
E67	MCROIR-17	94			
E68	MCROIR-18	95			
E69	MCROIR-19	96			
E70	MCROIR-20	97			
E71	MCROIR-21	98			
E72	MCROIR-22	99			
E73	MCROIR-23	100	0.0	Module shall pass all remaining tests	1-4

NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	E1-E73
Power supply current limit	3.82 A	NA	E1-E73

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:24Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Forcing voltage, (Vout)	0.0 V	± 50 mV	E1-E73
Pattern voltage high, (VH)	3.0 V	± 100 mV	E1-E73
low, (VL)	0.0 V	± 100 mV	E1-E73

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, no measurements are taken. Only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:24Z
Check the source to verify that this is the currentversion before use.